Supplementary Methods

S1 Source assets and grid record

Data-source roles and observation rules are reported in Table 1 and Sections 2.2–2.7 of the main manuscript. Exact annual Earth Engine assets are JRC/GSW1_4/YearlyHistory for 2000–2015, projects/global-surface-water/assets/GSW1_5/YearlyHistory_2016_2021 for corrected 2016–2021, and projects/global-surface-water/assets/GSW1_5/YearlyHistory_2022_2024 thereafter. Monthly assets are JRC/GSW1_4/MonthlyHistory through 2021 and projects/JRC/GSW1_5/MonthlyHistory_2022_2024 thereafter. The latter lacks system:time_start; genuine provider index/year/month metadata were used. Empty date-filter results were not dry observations.

Climate assets are UCSB-CHC/CHIRPS/V3/PENTAD and ECMWF/ERA5_LAND/MONTHLY_AGGR; radar and Dynamic World assets are COPERNICUS/S1_GRD and GOOGLE/DYNAMICWORLD/V1. The older monthly display used COPERNICUS/S2_SR_HARMONIZED. The later raw-value triage used archived Level-2A crop values and each product XML calibration, rather than reading display colors as reflectance.

The primary local raster grid preserves downloaded JRC alignment; climate intersections use EPSG:6933. The separate radar cohort grid uses EPSG:4326 with affine transform [0.00026949458523585647, 0, 0, 0, −0.00026949458523585647, 0]; its 10 m working grid is EPSG:32645. Categorical inputs use nearest-neighbor alignment and supported/water fractions use overlap-weighted aggregation. Reprojection does not add native detail or correct co-registration.

Boundary metadata are geoBoundaries gbOpen BGD ADM2, BGD-ADM2-16705992, represented year 2020, underlying Bangladesh Bureau of Statistics/OCHA ROAP, build 12 December 2023, source update 19 January 2023 and geoBoundaries commit 9469f09. The upstream license is CC BY 3.0 IGO. The original Nawabganj geometry is reported as Chapainawabganj.

S2 Revision identifiers and evidence ledger

The main manuscript reports the original and revised human counts and the assisted-review limitation. The single revised record is sample 144, opaque ID b_0a0e34935c6e74d9804cd2, scene S2B_MSIL2A_20230330T043709_R033_T45RXH_20240824T175150, acquired 30 March 2023. The reviewer changed nonwater to water after seeing rule output. The correction timestamp is revision provenance, not a new independent image inspection. Joins use sample_id plus scene_id. Original CSV/backup, corrected files and the cell-difference log remain in the code archive.

Supplementary_Data/table_manifest.json records row counts and SHA-256 values for 18 frozen main-result CSVs. Extended_Diagnostics/manifest.json identifies 20 additional fold-level, calibration and monthly diagnostic tables by original source. Rejected fits, support failures, unassigned labels and missing recovery durations remain visible. The 12 main figures are tied to original inputs through setup/inline_figure_manifest.json. The source ledger distinguishes registry metadata, abstract-only access and reviewed sections; accountable-author reference verification is still pending.

S3 Reproduction

Reproducibility.zip contains the completed code/results route and corrected active labels; the current run is code_only_final_20261016/analysis_v3_user_label144/. Earlier directories in that archive are historical. Reproduction also requires the genuine source folders: NW_Bangladesh_2000_2024, NW_Bangladesh_preprocessed, NW_Bangladesh_climate_baseline_1981_1999, NW_Bangladesh_automatic_water_2015_2024 and NW_Bangladesh_DW_sensitivity_2015_2024, plus dated raw crops and product metadata identified by source manifests. Large source rasters/crops are not duplicated here. Fresh Earth Engine acquisition requires an authenticated account and completed exports. Source licenses remain applicable.